

The Power Of Difference Assessment System

User Manual

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1.0 Introduction

TheSum is a nonprofit based in Charlottesville that has partnered with the Heather Heyer Foundation to form the Heyer Voices organization and is founded by J. Elliott Cisneros. The goal of TheSum is to stand in solidarity with all people, no matter their background. Through assessments, workshops, advocacy, and coaching, TheSum empowers people and organizations in solidarity across our differences. One key component of TheSum is the Power of Difference Assessment (PDA). This assessment identifies underlying biases in the thoughts of those who take it. Consultation is provided by either Mr. Cisneros or trained consultants to interpret these results. The improved web application developed for TheSum works to reduce the amount of manual work required to facilitate both the assessment and consultation process. The following manual will guide users through the additional features implemented within the system.

2.0 Overview

2.1 Functionality

TheSum Assessment System supports the following functions:

- User registration
 - Users can sign up using their email and will receive a verification link for confirmation
 - If email is already registered, you will need to use a different email to sign up for the PDA
 - Alternatively, if you receive permission to retake the PDA by the owner of TheSum, you will be asked to verify your account again using the email you originally registered with
 - Users can take some version of the PDA upon registration completion
- Dark Mode
 - Option to enable Dark Mode on the system
- Password reset
 - Admins and Consultants have the ability to reset their account passwords
- Supports 4 User Levels
 - Admin
 - Consultant Staff (Read Only)
 - View Only Admin (Read Only)
 - Normal users

- Power of Difference Assessment (PDA)
 - Users have the ability to take the assessment
 - PDF Generation
 - PDF of results will be generated upon completion and also emailed to the user
 - Exiting the Assessment Midway
 - Assessment sessions last for 1 hour upon starting the PDA
 - Users will return to the exact webpage and question they were on if they exit but return before the 1 hour period
 - Users will be unable to return to the PDA if they have exceeded the 1 hour time limit and will need to contact the system admin for permission to restart their assessment
 - URL Manipulation Security
 - Users who attempt to skip parts of the assessment by changing the URL will instead be redirected to the proper page.
 - 10to8 Scheduling
 - Users have the ability to sign up for a consultation time using the 10to8 widget upon completion of the PDA
 - User Accessibility
 - Users can take the assessment on various devices, such as mobile, desktop, and tablet.
- Admin Dashboard
 - Ability to view data at minimum if granted permission
 - Can edit, delete, and add data if given access
 - Ability to grant permissions to other accounts from the admin account
- Data visualization
 - Ability to view graphs of user information and responses
 - You must be logged in as an admin to see the data visualization
 - Graphs can be sorted by various categories of data
- Database manipulation with Amazon Web Services (AWS)
 - Ability to reset the database data and migrations
 - Ability to restore the database from snapshots

2.2 General Navigation

Upon accessing the system at <https://thepowerofdifference.org/>, you will be brought to the home page. You will be able to read the initial instructions and aspects of the PDA here as well as register to take the assessment.



The Power of Difference Assessment

The PDA may not be administered or reproduced in part or in whole, without the express written permission of The Sum - electronically or by other means.

IMPORTANT: In the reported results you will encounter information which is based on the Power of Difference Model and the PDA developers' experience. The Sum makes no claim as to the assessment's reliability and validity as this analysis is still in the beginning phases. If you are unprepared to encounter information of this nature, with which you may not agree, which you may find objectionable, and about which you may not be conscious, do not proceed.

The purpose of the PDA is to identify the unconscious patterns you "hold" or "value" in relationship to the intersection of power and our sociocultural differences of race, sexual orientation, disability, culture, religion, gender, and class. This information can help us to "leverage" or integrate the assets of these patterns while minimizing their limitations. The PDA has 70 items and requires approximately 20 minutes to complete.

[Register](#)

2.2.1 Power of Difference Assessment Walkthrough

After clicking on Register, you will be brought to the registration page, where you must enter your email, name, and phone number in order to register.

Register Here

Email

First name

Last name

Phone

Please enter valid US phone number

[Register](#)

If the admin has granted you permission to retake the assessment, please [click here to reverify](#)

After filling out the required information, a verification email will be sent to the one provided during registration. Check the inbox or spam folder of the email you provided for an email from info@thepowerofdifference.org. Once you have received it, click on the hyperlink prompted in the email to finish verifying your registration.



info@thepowerofdifference.org
to me ▾

6:28 PM (1 minute ago) ☆ ↶ ⋮

Hi Anonymous,

Thank you for registering to take the Power of Difference Assessment. [Please click here to verify your email.](#)

NOTE: Once you click the link and start the assessment, you must finish it within an hour. You will not be able to change devices while you are taking the assessment.

We hope you find the PDA meaningful and helpful!

The Sum

Charlottesville, VA

www.thesum.org

Please send any replies to info@thesum.org

Once verification is complete, the hyperlink from the email will open a new webpage where you are given two options. If you have an access code, you may use it to gain entry to take the PDA. Otherwise, you must make a one time payment in order to gain paid access to the PDA.

Institutional Access	Paid Access
If your institution provided you with an access code, please enter it here.	If you want to take the PDA and receive an hour-long results consultation (on the phone or in-person), please click the button below to pay through PayPal. On the PayPal site, use the gray button at the bottom that says pay with debit or credit card. The cost is \$150.00, though it is available on a sliding scale.
Enter Access Code	\$150.00
Verify Code	Buy Now

After you successfully enter the PDA, you will be prompted to fill out personal demographics questions. Upon completion, you will be taken to the beginning of the PDA where instructions will be displayed on how to take the PDA.

Power of Difference Assessment - Demographics

Please pay close attention to the 19 demographics questions as they not only assist with our research and help us to improve the survey, in certain cases, your score is automatically adjusted according to your identified demographic location.

Age

Current religious/non-religious identification

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Once you have submitted all 70 responses, you will be taken to a final page prompting you to sign up for a consultation time slot. At the same time, an email containing a PDF report of your results will be generated and sent to you. After signing up for a consultation, you, as the user, are finished with the PDA and can await followup from Consultants.

Congratulations!
You have completed the PDA.

[Reset for next user](#)

You have selected the free version of this assessment. Use the slots below to sign up for a free 10 minute consultation.

Service

Where

When

Details

March 2020 ▾

TODAY FIRST SLOT << < > >>

FREE (or desired donation) 10 minute PDA consultation for research study.

Elliott Cisneros

Online / Phone

Times are displayed in EDT (-04:00) America/New_York

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Dark Mode

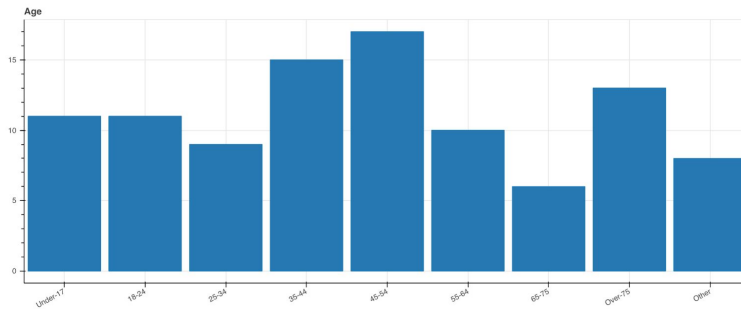
2.2.2 A Guide to the Data Visualizations

Data visualizations are available to admin at <https://thepowerofdifference.org/visualization/>

The first type of visualizations breaks down the data by demographic. The graph displays the total number of assessments taken by each category of the demographic. For example, 'Age' is the default demographic, and the graph displays the number of PDA participants in each age group. The dropdown can be used to select different demographic categories.

Demographics

Age

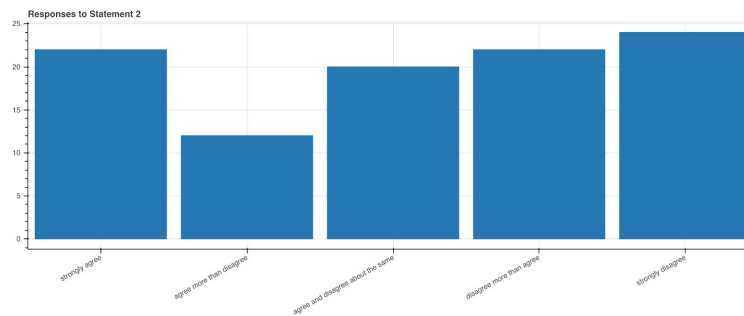


The next type of visualization displays the data by responses to each statement. The response index lists all of the statements in the PDA. Selecting a specific statement displays a graph representing the number of users who selected each response.

Responses

2. Some religions/worldviews are really violent and destructive underneath.

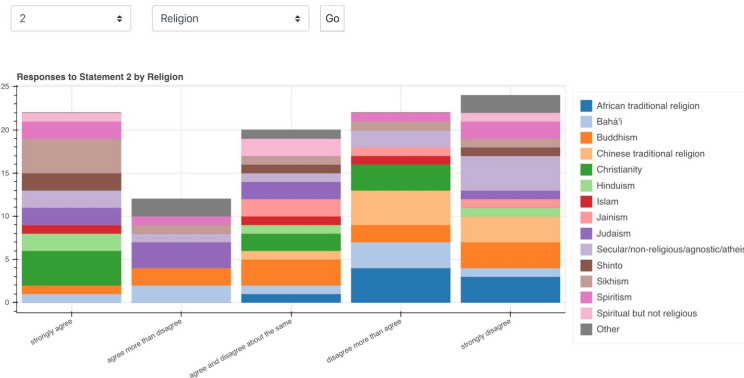
2 None



By using the second dropdown box (the first will select a new statement number), the graph can be colorized to show the distribution of responses across a specific demographic category, for example religion.

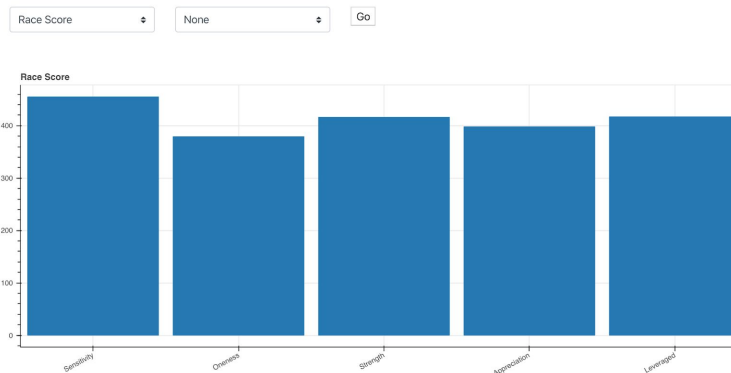
Responses

2. Some religions/worldviews are really violent and destructive underneath.



Finally, score visualizations display the score in each category (sensitivity, oneness, strength, appreciation, leveraged), totaled across all of the completed PDAs. Overall scores, as well as scores for gender, race, religion, sexual orientation, disability, culture, and class can be viewed using the first dropdown. =

Scores



Similarly to the response visualizations, the second dropdown will colorize these scores by the demographic information of the assessment takers. The graph below shows the race score, similar to the one found on the second to last page of the assessment results pdf, totalled across individuals who have taken the exam, and color coded by the disability demographics of the users.

Scores



2.3 User Levels

There are four roles that are assigned: Admin, Consultant, Organization, and User.

2.3.1 Admin

- Has all permissions across the admin dashboard
 - Has ability to view, edit, add, and delete all data involving:
 - Verified users, unverified users, demographics, gender scores, race scores, religion scores, sexual orientation scores, disability scores, culture scores, class scores, overall scores, assessments, responses, access codes, and consultants
 - Has ability to download PDFs from each user
 - Has ability to create Consultant Staff logins using email verification
 - Can assign specific data to be viewable by Consultants
 - Can remove Consultant accounts
 - Has ability to create and delete Organization accounts
 - Has ability to create new admins with equal levels of access or less
 - Has ability to allow users to retake the PDA
 - Has ability to create access codes for the PDA
 - Can set the number of uses for each access code created
 - Has ability to change all account passwords
 - Can search and sort through data

2.3.2 Consultants

- Can view all the data on the admin dashboard that is assigned to them by the admin
 - Data is limited to information categorized under Assessments, Demographics, and Users
- Can change their own account password
- Can search and sort through data

2.3.3 View-Only Admins

- Has the same level of access on the admin dashboard as the Admin except that it is only granted view permissions
 - Cannot edit, add, or delete data

2.3.4 Users

- Basic role assigned to all people who take the PDA
 - Can sign up for the PDA
 - Can take the PDA
 - Can receive generated results by email
- Can select and schedule a consultation with a Consultant

3.0 Description of Functionality

3.1 Creating A Super User

- If you ever try to use “eb -----” in the terminal, make sure you have your profile set properly (see installation instructions... you may have to add --profile <name of profile> to the command). If it gives you an error to run eb init first, do that (and set your profile properly there too!)

3.1.1 What if I Lost my ssh Keys?

- Go to a terminal that is inside the src directory for the project.
 - Run eb init -i
 - Follow the instructions
 - When prompted, make new ssh keys
 - You may use a default name or give it a new name
 - Go to your elastic beanstalk environment on the AWS console

- Under configuration, modify security, set the EC2 key pair to the one that you created
 - After applying, your elastic beanstalk system now uses those ssh keys!
- Note: the ssh keys are stored in ~/.ssh on your computer
 - The one with no extension is your private key and the one with an extension is the public key
 - Only you know your private key!
 - Example: key = private key and key.pub = public key

3.1.2 ssh in and Create A New Super User

- Run “eb ssh <environment name>”
 - If this does not work, you may have to run “eb init” first, select the matching region and environment for your application
- If asked to approve the host, say yes (it should tell you what to type for yes)
- Once you have ssh access, you will want to make a new super admin
- Run `source /opt/python/run/venv/bin/activate && source /opt/python/current/env && cd /opt/python/current/app/ && python manage.py createsuperuser`
- Follow the steps here!
- When done, type exit and press enter to close the connection. After this, you can close your terminal.

3.1.3 Deleting a Super User

- Run “eb ssh <environment name>”
 - If this does not work, you may have to run “eb init” first, select the matching region and environment for your application
- If asked to approve the host, say yes (it should tell you what to type for yes)
- Once you have ssh access, you will want to see which super users exist
- Run “source /opt/python/run/venv/bin/activate && source /opt/python/current/env && cd /opt/python/current/app/ && python manage.py shell”
- Next, run “from django.contrib.auth.models import User”
- To view the current super users, run “User.objects.all().filter(is_superuser=True)”
- You will see a list formatted like so: [<User: username>, repeated until the end of the list] Where a username is after the semicolon, but before the ending >. In the example, username is the username.
- To delete a specific super user, run “User.objects.get(username=“NAME”, is_superuser=True).delete()” where NAME is replaced with the username of the super user that you want to delete

- When done, type `exit()` and press enter to close the python shell. Next, type `exit` and press enter to close the connection to AWS. After this, you can close your terminal.

3.2 How to Upload a New Version of The Code

- If you ever try to use “`eb -----`” in the terminal, make sure you have your profile set properly (see installation instructions... you may have to add `--profile <name of profile>` to the command). If it gives you an error to run `eb init` first, do that (and set your profile properly there too!)
- To do this, you need to open a terminal and navigate to the location of your source code. You will need to be in the same folder where `manage.py` is. In the case of our GitHub repo, this means you will need to be inside the `src` directory.
- Make sure you have the `questions.json` file in here! It may not be there if you just downloaded the code from our GitHub repo, so make sure to add it if it is not there.
- Run “`eb deploy`”
 - You may have to run “`eb init`” first. If you have to do so, you will be told by `eb deploy`. Make sure you choose the right region and environment in `eb init`.

3.3 Database Manipulation

3.3.1 Database Reset

If you run into a case where the database needs to be reset or issues with migrations arise, please consult the following instructions for rebuilding the environment.

First, take a snapshot of your database on Amazon Web Services (AWS) before resetting in order to avoid losing data.

- A snapshot can be taken on the RDS page corresponding to the database you are resetting. Follow steps 15-18 from the Installation Instructions guide to get the management console for your database. Click on “Actions” and select “Take snapshot” from the dropdown options.

RDS > Databases > aa171f1bsk4ulwq

aa171f1bsk4ulwq

Summary

DB identifier aa171f1bsk4ulwq	CPU 3.05%	Info Available	Class db.t2.micro
Role Instance	Current activity 0 Connections	Engine PostgreSQL	Region & AZ us-east-1f

Modify

Actions

- Stop
- Reboot
- Delete
- Create read replica
- Promote
- Take snapshot
- Restore to point in time

You will want to restore a database from the snapshot you just took so that you can still access the data. In order to do so, follow the guide here:

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_RestoreFromSnapshots.html

Afterwards, add security rules in order for pgAdmin (a tool you will use to connect to the database) to connect to the database.

- You can get to the security group by going to the restored database's page in the RDS section of the AWS console. Scroll down and click on VPC security group in the Security section.

RDS > Databases > aa171f1bsk4ulwq

aa171f1bsk4ulwq

Summary

DB identifier aa171f1bsk4ulwq	CPU 3.17%	Info Available	Class db.t2.micro
Role Instance	Current activity 0 Connections	Engine PostgreSQL	Region & AZ us-east-1f

Connectivity & security

Monitoring

Logs & events

Configuration

Maintenance & backups

Tags

Connectivity & security

Endpoint & port Endpoint aa171f1bsk4ulwq.cwjhifzdgx2s.us-east-1.rds.amazonaws.com Port 5432	Networking Availability zone us-east-1f VPC vpc-a1f9d5db	Security VPC security groups rdc-awseb-e-75bwafd2pj-stack-awsebrdsdbsecuritygroup-1kwwy4o15g47-7wn1 (sg-061b60a84fb7c30c0) (active)
--	---	--

EC2 > Security Groups

Security Groups (1/1) Info

Filter security groups

search: sg-061b60a84fb7c30c0

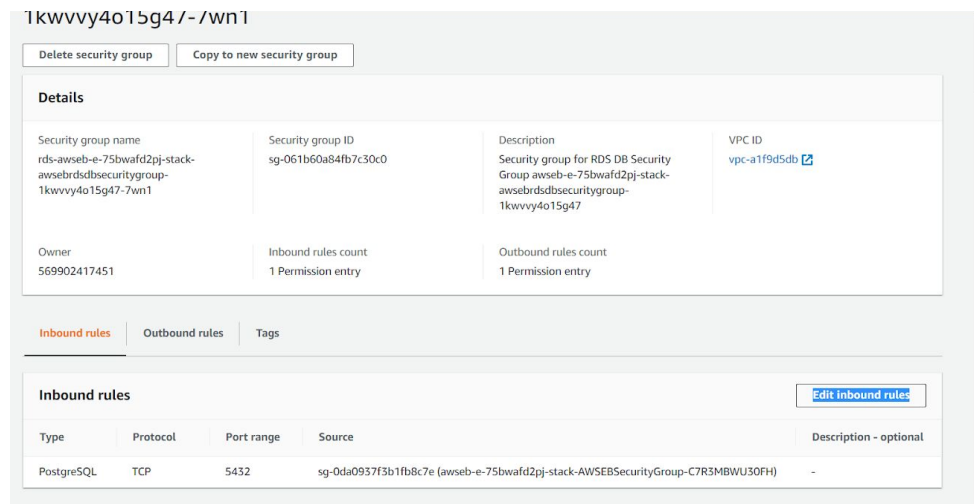
Clear filters

Actions

Create security group

Security group ID	Security group name	VPC ID	Description	Owner	Inbound
sg-061b60a84fb7c30c0	rdc-awseb-e-75bwafd2pj-stack-awsebrdsdbsecuritygroup-1kwwy4o15g47-7wn1	vpc-a1f9d5db	Security group for RDS...	569902417451	1 Permis

Click on the name of the security group and then scroll down until you find Inbound Rules. Click on “Edit Inbound Rules” once you locate it.



Add a rule for HTTP and select Anywhere for where you want traffic to be from (you may optionally fill out the description)

Once you have added the rule for HTTP, save the changes. This will add the 2 new rules that are shown above and will allow the pgAdmin program to connect to your restored database.

- Note: The previous steps cannot be done on a database that is attached to an elastic beanstalk environment. If that is the case for you, take a snapshot as stated earlier and make a database from the saved snapshot in order to perform the previous steps.

Afterwards, you will want to modify the security group for your database. Once you are finished browsing with pgAdmin, remove the 0.0.0.0/0 rule and add it back the next time you want to browse the database.

- Note: This step is included for security purposes.

3.3.2 Connecting to the Restored Database

In order to connect to the restored database, you will need to use pgAdmin. You can download pgAdmin at: <https://www.pgadmin.org/download/>.

- Follow this guide in order to connect to your database from pgAdmin: https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_GettingStarted.CreatingConnecting.PostgreSQL.html#CHAP_GettingStarted.Connecting.PostgreSQL
 - If you do not remember the username or password for the database, you should be able to find them in the environment variable configurations for your environment under RDS_USER and PRODUCTION_DB_PASSWORD.

3.3.3 Rebuilding and Redeploying the Environment

- Rebuild environment and redeploy
 - To rebuild the environment, on the elastic beanstalk page for your environment, click on actions and then click on rebuild environment.
 - Rebuilding the environment will also reset the database (which is why the steps above have you make a snapshot). You will not be able to continue using that old database with the rebuilt environment, but the steps above tell you how you can still access the old data.
 - After rebuilding is done, you will have to deploy your code to the environment again (see the installation instructions if you have forgotten how to do this)
 - **IMPORTANT: Since you have a new database, you will have a new RDS_HOST and RDS_DB_NAME!** In order to find what their new values should be, you can follow step 18 of the installation instruction. You will have to set these environment variables in your rebuilt environment. See steps 26 through 30 of the installation instructions if you need to remember how to do this.
 - Redeploy the app!
 - If you receive WSGI issues, please restart the app servers and redeploy (see the installation instructions if you have forgotten how to do this)
 - You will have to make a new super admin since you now have a new version of the database. All consultants and view only admins will also have to be added back into the system.

- You will most likely have to redo the steps to setup HTTPS in order to get HTTPS to work again for your rebuilt environment (see installation instructions). This will likely include changing the URL for your alias record in route 53 (see instruction installations) based on the URL for your new environment. You will also want to delete all old environments after this is done.
 - For these purposes, we have an old configuration already saved. It is PDA-feb-26-2020. It is missing some required environment variables (such as FROM_EMAIL), but it can be used as a base and those environment variables can be added back in. Please see the installation instructions for all needed environment variables. When doing this, remember that you will have to find your load balancer for the environment in elastic beanstalk and add in HTTPS rules. You will also have to find the load balancer in the AWS EC2 console and set HTTPS redirection rules.

3.4 Email Functionalities

3.4.1 Viewing Emails

While not often, there is sometimes a need to check the emails that the AWS account is receiving.

- Open the AWS Console and navigate to Amazon S3 under storage in Amazon Services
- Select Buckets
- Click on the Bucket named pda-email-receiver (If this isn't there refer to the installation instructions on setting emails on AWS, the name should be something similar to this)
- On the next page, the website will redirect the page to an overview of the bucket.
- In this overview tab, the emails are stored as blank file types, with messy names.
- To view an email click on one of these files and hit download.
- With the file downloaded open the file with a text viewer (such as notepad) to view the email.

3.4.2 Adding and Verifying Domains

If you wish to change the domain of the application, you will have to add and reverify it in Amazon Simple Email Services (SES). Assuming the new domain name has already been selected:

- Open the AWS Console and navigate to Amazon Simple Email Service (SES).
- Under identity management, select the “Domains” option.
- Select “Verify a New Domain” at the top.
- Type the name of the new domain, and check the box for generating DKIM settings.
- Select “Verify This Domain.”
- Create or view an Amazon S3 bucket for receiving emails. Instructions for how to do this can be seen in part 90 of the installation instructions.
- Find the verification email that was sent to the domain. Refer to section 3.4.1 on how to view emails.
- Copy paste the verification link into a browser and access that page to verify the domain.

3.4.3 Adding and Verifying Email Addresses

- Open the AWS Console and navigate to Amazon Simple Email Service (SES).
- Under identity management, select the “Email Addresses” option.
- Select “Verify a New Email Address”
- Enter the email address that you wish to send from. For example, this could be “info@thepowerofdifference.org”
- Go to Amazon S3 and find the S3 bucket that was created for receiving emails.
- Find an email that was recently received in the bucket, which should be the verification email.
- Open the verification email to verify the email address. For more help on how to view emails in the S3 bucket, refer to section 3.4.1.

3.4.4 Sending Test Emails

When the email addresses and domains have been set up, it is important to ensure that they work.

3.4.4.1 Sending Test Emails through Email Addresses

- Open the AWS console and navigate to Amazon Simple Email Services (SES).
- Under identity management, select the “Email Addresses” option.
- Select the email address that needs to be tested by checking the box to the left of the name.
- Click on “Send a Test Email,” which will open up an email prompt
- On the prompt, it will tell you to fill out the To, Subject, and Message sections of the email.
- In the To section, put an email address that you already have verified.

- For the Subject, and Message it does not matter what you put in there.
- Click on Send test email to send the message
- Go to the account that you sent the test email and verify it's there.
- It is recommended you do this again with the recipient being an unverified email account. If the unverified email is not received, refer to Section 93 of the installation instructions to make sure the account is moved out of sandbox mode.

3.4.4.2 Sending Test Emails through Domains

- Open the AWS console and navigate to Amazon Simple Email Services (SES).
- Under identity management, select the “Domains” option.
- Select the domain that needs to be tested by checking the box to the left of the name.
- Click on “Send a Test Email,” which will open up an email prompt
- On the prompt, it will tell you to fill out the From, To, Subject, and Body sections of the email.
- The contents of From, Subject, and Body do not matter.
- For To, select an email address that is readily accessible to view the email that will be sent.
- When all fields have been filled out, click on “Send Test Email”
- Open the inbox of the email that was used in the “To” section to view the email.

3.4.5 Changing FROM_EMAIL and BCC_EMAIL Environment Variables

If you ever wish to change the email from “info@thepowerofdifference.org” to another email, or change the email that is BCC’d when PDFs are sent, there is a simple way to do this.

- Open AWS Console and navigate to Elastic Beanstalk.
- Select the “env-pda” environment in the center panel.
- Select “Configuration” on the left panel underneath “env-pda”
- The configuration overview will display with many categories. Select “edit” under the “Software” category.
- Scroll down until the “Environment properties” section.
- Modify “BCC_EMAIL” and “FROM_EMAIL” to be their new values.
- Apply the changes by clicking on the “Apply” button at the bottom.

Note: If the “FROM_EMAIL” variable is changed, the new email will have to be verified. Refer to section 3.4.3 on how to verify a new email.

3.5 Admin Functionalities

3.5.1 Data Manipulation

The admin has the ability to view, edit, add, and delete any data on the admin dashboard.

Core		
Access codes	+ Add	Change
Assessments	+ Add	Change
Class_scores	+ Add	Change
Consultants	+ Add	Change
Culture_scores	+ Add	Change
Demographics	+ Add	Change
Disability_scores	+ Add	Change
Gender_scores	+ Add	Change
Race_scores	+ Add	Change
Religion_scores	+ Add	Change
Responses	+ Add	Change
Scores	+ Add	Change
Sexual_orientation_scores	+ Add	Change
Statements	+ Add	Change
Unverified users	+ Add	Change
Users	+ Add	Change

To add something to each category above, simply click on “Add” and you will be prompted to enter a new entry into that category. Similarly, clicking on “Change” will pull up a list of all the data entries under a category and you will have the option to edit any entry within the list.

- Clicking on a category and then clicking on an entry afterward will achieve the same result if you are trying to edit an entry.

On the side of each individual entry, you will see buttons that include “Save” and “Delete”. When making edits, you must click on “Save” in order for the changes to be saved. If you were to delete an individual entry, you can click on “Delete”.

Authentication and Authorization
Core
Ipn
Pdt

Access codes
Assessments
Class_scores
Consultants
Culture_scores
Demographics
Disability_scores
Gender_scores
Race_scores
Religion_scores
Responses

Scores
Sexual_orientation_scores
Statements
Unverified users
Users

Home > Core > Class_scores > Class_Score object (1)

Sensitivity: *
2

Oneness: *
2

Strength: *
3

Appreciation: *
3

Leveraged: *
3

Score: *
Score object (1)

Save

Save and add another

Save and continue editing

Delete

History

Any changes, additions, or deletions will be logged in the system history and can be viewed on the side of the main dashboard under “My Actions”.

My actions		
	Statement 1	Statement
	Elliott - Uses Left: 1	Access code
	Elliott - Uses Left: 1	Access code
	Elliott - Uses Left: 1	Access code
	jelliottcisneros	User
	Elliott - Uses Left: 1	Access code
	jelliottcisneros	User
	jelliottcisneros	User
	Elliott - Uses Left: 5	Access code

3.5.2 Setting up Access Codes

Under the access code category, you will be able to set up access codes for users that take the assessment using the Access Code option.

- The name is used to easily identify the access code you created.
- The actual code can be anything you want.
- The number of uses is unlimited if the value set in the “Uses left” box is -1. Otherwise, please enter a positive number greater than 0 and each time the code is used, the number will decrease by 1, until it reaches 0. You can view each access code you create to see the remaining number of uses at any time.

Home > Core > Access codes > Add access code

Name:	Example Access Code
	Nickname for identification purposes.
Code: *	CodeTest
Uses left: *	1
	Enter -1 for unlimited use

3.5.3 PDF Access

Under “Assessments”, for each entry you will be able to see various information about the user that took it as well as having the option to download the PDF of their results.

Home > Core > Assessments > Assessment object (4)

User:	John Cisneros (jelliottcisneros@thesum.org)   
Access type: *	Institution
Email: *	jelliottcisneros@thesum.org
Date started: *	2020-03-30  Today
	Note: You are 4 hours behind server time.
Last Statement: *	0
PDF:	Currently: pdfs/jelliottcisneros@thesum.org_4_results.pdf  Clear Change: Choose File No file chosen
Consultants:	 Hold down "Control", or "Command" on a Mac, to select more than one.
More Details:	Click here

If you click on the link provided next to “More Details”, you will be taken to a more detailed view of the user information, where it will display their demographics, responses, and a link to the PDF that will open in your browser window.

[←Go back to assesment lists](#)

Assessment 4 Details

User	Demographics	Responses	Other
Consultants		No consultants assigned	
PDF Report		Click to view	

3.5.4 Retaking the Assessment

Users are only able to retake the assessment if given permission by the admin. In order to give permission, you must first select a user under Users on the admin dashboard.

Email: *	example@example.com
First name: *	Anonymous
Last name: *	Anonymous
Phone: *	1234567890 Please enter valid US phone number
Can retake	☒

Make sure that the box next to “Can retake” is checked. This will allow users to take the assessment again. When taking the assessment again, users must use the original email they signed up with and go to the registration page. They must fill out the registration form again as if they are taking the assessment for the first time. After completing this, a new verification email will be sent to the user that will allow them to verify their information and take a new assessment.

3.5.5 Creating Consultant Accounts

Admin selects the Consultant link, bringing up the Consultants data page. From this page, the Admin will select the “Add consultant” button

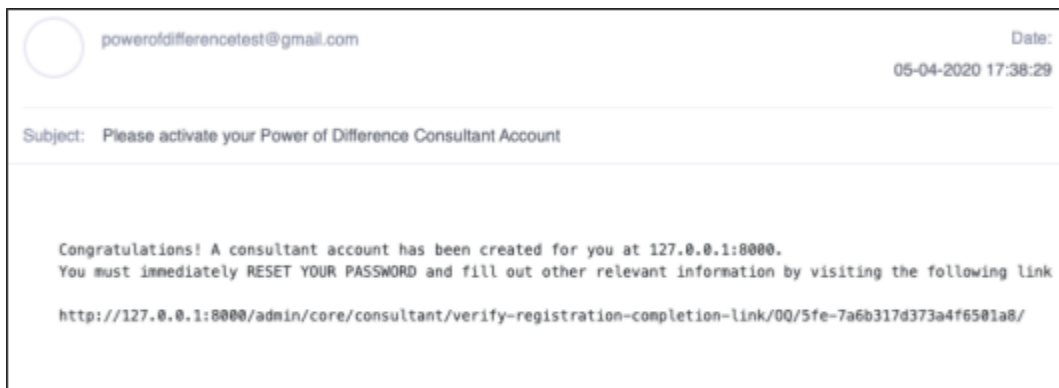
The screenshot shows the Admin Dashboard with a sidebar on the left containing a 'Core' section and a main content area. The sidebar has a 'Core' tab selected. The main content area has a navigation bar with tabs: 'Access codes', 'Assessments', 'Class_scores', 'Consultants' (which is underlined), 'Culture_scores', 'Demographics', 'Disability_scores', 'Gender_scores', 'Race_scores', 'Religion_scores', 'Responses', 'Scores', 'Sexual_orientation_scores', 'Statements', 'Unverified users', 'Users', and 'View Only Admins'. Below the navigation bar, there is a search bar with a 'Search' button and a green 'Add consultant' button.

The Admin will then enter the email address for the Consultant user and click “Save”

Home > Core > Consultants > Add consultant

Email address:

Now, the Consultant user will receive an email on their end from the Power of Difference Assessment, notifying the user that an account has been created for them and they need to reset their password with the link provided.



After entering the requested account information from the link, Consultant users can now log into their admin site and view the data of those customer users that they're assigned to.

3.5.6 Assigning a Consultant to an Assessment

The Admin selects the "Assessment" link and chooses a specific Assessment to view/edit.

Core

Access codes Assessments Class_scores Consultants Culture_scores Demographics Disability_scores Gender_scores Race_scores Religion_scores

Responses Scores Sexual_orientation_scores Statements Unverified users Users View Only Admins

Search Add assessment

Assessment	User	Access type	Email	Demographic	Date started	Scores	Last Statement	Recalculate Scores
Assessment object (2)	assessment2 example2 (assessment2@example.com)	Free	ninetails345@upcmail.com	Demographic object (2)	March 14, 2020	Score object (2)	0	Score object (2)
Assessment object (1)	jace jupiter (jicecec431@provamail.com)	Free	jicecec431@provamail.com	Demographic object (1)	March 14, 2020	Score object (1)	0	Score object (1)

Action: 0 of 2 selected Go

2 assessments

Once the Admin chooses an Assessment, the Admin can assign consultants to by selecting specific registered Consultant account(s) from the drop-down menu while holding down control(windows)/command(mac) and click “Save”

Core

Access codes Assessments Class_scores Consultants Culture_scores Demographics Disability_scores Gender_scores Race_scores Religion_scores

Responses Scores Sexual_orientation_scores Statements Unverified users Users View Only Admins

Home > Core > Assessments > Assessment object (1)

User: jace jupiter (jicecec431@provamail.com)

Access type: * Free

Email: * jicecec431@provamail.com

Date started: * 2020-03-14 Today

Note: You are 4 hours behind server time.

Last Statement: * 0

PDF: Currently: pdfs/jicecec431provamail.com_1_results.pdf Clear

Change: Choose File No file chosen

Consultants: consultant-example

Hold down "Control", or "Command" on a Mac, to select more than one.

Save

Save and add another

Save and continue editing

Delete

Object permissions

History

Now, the selected Consultant account(s) has the permission to view the responses, demographic data, and other relevant information of the user who took the specific Assessment.

3.5.7 Creating View-Only Admin Accounts

The process for creating View-Only Admins will follow the same steps as Consultants. However, instead of choosing the “Consultants” link, the Admin will choose the “View Only Admin” link and start the registration process for View-Only Admins by clicking the respective “Add View Only Admin” button.

3.5.8 Changing Statements

Although every other entry in the database can be changed, the Statement entries require a little more care.

- Changing the number field is highly discouraged. If it becomes necessary to change the ordering, it should be ensured that the Number is between 1 and 70. And no two statements share the same number.
- There are quite a few statements for which the power perspective depends on the demographic of the user who took it. In this case, first set what the alternate power perspective is, and then choose the related demographic field. Then choose the demographic option for which the alternate power perspective should be used, instead of the Primary power perspective.

Home > Core > Statements > Statement 24

Number: *	24
Title: *	Most people of color are smarter than most white people.
Sociocultural location: *	Race
Primary power perspective: *	Appreciation
Secondary power perspective:	Strength If power perspective is independent of demographics, leave blank. Otherwise, first set the related demographic field on which the power perspective will depend on. Then set the single demographic choice for which power perspective should be set to secondary power perspective instead of primary power perspective. Then set the secondary power perspective
Related Demographic Field:	Race (as perceived by others) Not required if secondary power perspective is not set
Field choice that should lead to secondary power perspective instead of primary power perspective being set:	As a Person of Color Not required if secondary power perspective is not set

- **Most importantly**, please also update the questions.json file that came with your codebase, so that future code updates will include the changes you have introduced. It uses a file format called [JSON](#). The easiest way to do it you'd be to
 - SSH into the environment. (See steps 1 & 2 of 3.1.2 for more details)
 - Copy `source /opt/python/run/venv/bin/activate && source /opt/python/current/env && cd /opt/python/current/app/ && python manage.py dumpdata --indent 4 core.question` paste it onto the terminal and press enter.
 - Copy paste the 70 statements (starts with a "[" and ends with a "]") into the questions.json file you have.

3.6 Changing the PDF Report

Making changes to the PDF requires directly manipulating the source code, which can be a little daunting. However, making small aesthetic changes does not require a high-level technical understanding. The library that we use to automatically generate the PDF Report is called ReportLab, and their documentation can be found here:

<https://www.reportlab.com/docs/reportlab-userguide.pdf>

We do not use all of the services offered by this platform, but the documentation breaks down many of the concepts used in our code.

3.6.1 Finding the Code

From the home directory of the project, navigate to the “src” folder, then “core”, then “views.py”. This is where the programmatic parts of the application occur. Use the Ctrl-F function to find “PDF generation starts here:”. Between this line and the line that says “PDF generation ends here” is where the PDF is programmatically created.

The code that generates each page is preceded by a line that says “page [#]”, where “[#]” is replaced with the number of the page.

3.6.2 Changing the Page Layout

Each page starts with a line that looks like `p.translate([#] * inch, [#] * inch)`. Changing the values multiplied by inch will change the margins on the page. Be careful when doing this though, as this may affect the placement of other objects on the page.

To change the orientation of a page, change the line that reads `p.setPageSize(portrait(letter))` to read `p.setPageSize(landscape(letter))`. To reverse this change, change it back to the original.

When a page is finished and ready to be added to the report, add the line `p.showPage()` to the very end of the page.

3.6.3 Adding or Changing an Image

Images that appear in the PDF report are saved in the `src/core/static/core/img` directory. This is important, because otherwise the program will not be able to find the image to insert it into the PDF. The ReportLab library is only able to handle image types in the .JPG format, inserting any other type of image may result in an error. To add the image to a page, write the following line:

```
p.drawImage(static("core/img/[image name]")[1:], [x location], [y location], [width], [height])
```

Replace “[image name]” with the name of the image, including the “.jpg” file extension. Replace “[x location]” and “[y location]” with the location you want the image placed on the page, relative to the bottom left corner. For example, using “1*inch” as the x location and 1*inch as the y location will place the image 1 inch to the right and 1 inch up from the bottom left corner of the page. The program will scale the image appropriately to fit in the width and height settings, so using a ratio of values that is not the same as the image’s original ratio will cause the image to stretch or shrink.

Images that already exist in the PDF report can be edited by adjusting these values as well. Images should be labeled in the code with the line “# Image: ” followed by a description.

3.6.4 Editing Text

Text on the page is rendered in two ways. The first is using the `p.drawString([x location], [y location], “[text]”)` method. This will write text on the page in a single line at the location specified. This has several drawbacks, one being that the text will show as a single line, even if it runs off the page, so this should only be used for short lines of text. The second is that there is a small number of formatting changes that can be made. However, you can change the font being used in one of these methods, using the `p.setFont(“[font name]”, [font size])` method. This will change the font so that all subsequent uses of `p.drawString` write in the specified font name and size.

For larger blocks of text, a more complicated method is used. This method involves creating a style object, then creating a Paragraph object. To create a style object, write the following line of code:

```
pstyle = ParagraphStyle('default')
```

This starts with a default style for a paragraph. To make changes to the default paragraph style, write the following block of code:

```
[New style name] = ParagraphStyle(
    '[style name]',
    parent=pstyle,
    [style change 1]=[new style 1],
    [style change 2]=[new style 2],
    ...
)
```

This code creates a new style, called whatever you put in the [style name] block, and uses the changes listed in the object. For example, setting [style change 1] to “fontSize” and [new style 1] to “16” will change the font size to 16. There are a variety of other styles that can be changed, the codes for which can be found in the ReportLab documentation linked above. Once the paragraph style is set, add this line:

```
par = Paragraph("[your text here]", [style name])
par.wrapOn(p, [width], [height])
par.drawOn(p, [x location], [y location])
```

This will write the text in [your text here] on the page at [x location], [y location]. On page 1, multiple Paragraphs are used as part of a ListFlowable. This ListFlowable object creates a bulleted list, with each Paragraph representing a bulleted item.

Using Paragraph objects also allows a bit easier formatting to be done to the text. The [your text here] field can contain HTML tags. For example, to make text italic, bold, or underline, surround the text with tags like shown:

```
<i>Italic text</i> ---> will print on report like ---> Italic text
<u>Underline text</u> ---> will print on report like ---> Underline text
<b>Bold text</b> ---> will print on report like ---> Bold text
<i><u>Underlined and italicized text</u></i> ---> will print on report like
---> Underlined and italicized text
```

3.6.5 Tables

Tables work similarly to how the Paragraph objects mentioned above do. They require a data object, a TableStyle object, and a few extra parameters. The first step to creating a table is defining the data. Table data is represented as a doubly-nested list (that is, a list of lists). The outer list represents rows, and the inner lists represent columns. For example, the following line represents five rows, each with one column:

```
data = [[[""], [""], [""], [""], [""]]]
```

The following line represents 3 rows, with 3 columns each:

```
data = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
```

And is represented on the report like this:

1	2	3
4	5	6
7	8	9

Once the data is collected, the table needs to be styled. To do this, we need to create a `TableStyle` object. `TableStyle` objects are a list of style adjustments to be made. Like a `ParagraphStyle`, there are several attributes that can be changed, and they are listed in more depth in the [ReportLab documentation](#) linked above. In general, a `TableStyle` object should look like this:

```
[style name] = TableStyle([
    ('[style 1]', ([x1], [y1]), ([x2], [y2])),
    ('[style 2]', ([x1], [y1]), ([x2], [y2])),
    ...
])
```

Where the `[style name]` represents a name for the style (it can be anything, but you will reference it later when applying the style), `[style 1]` and `[style 2]` represent the styles that will be changed (a list of relevant styles can be found in the [ReportLab documentation](#)), and `[x1]`, `[y1]`, `[x2]`, and `[y2]` represent the coordinates relative to the top left corner of the table that you want the style applied to. For example, `('SPAN', (0, 0), (2, 0))` will merge the cells from the top-left cell to the cell two spaces to its right.

Once the data is collected and the style is defined, the table can be printed. This is done as follows:

```
[table name] = Table([table data], colWidths=[column width],
rowHeights=[row heights])
[table name].setStyle([style name])
[table name].wrapOn(p, [width], [height])
[table name].drawOn(p, [x location], [y location])
```

The [table name] variable can be anything, but the two must match and may be referenced later. The [table data] is the name of the data that we defined earlier. [column width] and [row heights] can be represented as a value multiplied by the `inch` variable, like we did earlier, as can the [width], [height], [x location], and [y location] variables, which refer to the width and height of the table, and the x and y locations of the table on the page. Finally, the [style name] variable is the same as the style name that we defined above.

Tables can also programmatically add to their styles and data. For example, when populating the tables on page 3 of the PDA report, we programmatically fetch the score data from the database, then add the data to the table. This is done with the `[data name].append([new data])` method, where the [new data] represents a new row being written to the table, in the same syntax as described above. Styles can be added with the `[style name].add('[style]', ([x1], [y1]), ([x2], [y2]), [value])`. This will add a new style on top of the styles originally defined in the TableStyle object named [style name]. In our application, this method is used when adding data from the database to the table, and to programmatically change the background color of cells based on which response is chosen.

While making changes to the styles we've already applied may be useful, I would recommend against changing the sections where data is fetched from the database, as this can cause the application to fail if not done properly.

3.6.6 Graphs

Graphs in ReportLab are similar to tables. To facilitate the creation of many bar graphs on page 5, we've added two "helper methods" called `generate_total_graph()` and `generate_graph()`. To find these in the file, use the Ctrl-F function to search for "def generate_total_graph()". These functions take care of a lot of the setup for the graphs. To make changes to any of the formatting of the table, change the values in these methods.

The rest of the graph data is set under the "# page 5 - graphs" heading. This includes the title over graph and its formatting, as well as the placement of the graphs. Graphs in ReportLab are added to Drawing objects, which specify the length and height of the object to be added to the PDF document. Changes to the variables setting the placement of the graph set its placement within the drawing object. Once the graph is finished, the graph is added to the drawing using the `[drawing name].add([graph_name])` method, and the drawing is added to the page using the `renderPDF.draw([drawing name], p, [x location], [y location], showBoundary=[True or False])` method, where [x location] and [y location] refers to the location of the drawing on the page.

The graphs on page 6 do not use the same helper methods, and are instead generated under the “# page 6: last page” header, but their attributes are very similar. Data is set in the [graph name].data = [...] method, and their attributes are set after.

3.7 Error handling

[Website errors](#) could be of many types. Of this we are only responsible for 500 errors. Our current approach to error handling is,

- When a user comes across an error they are shown the 500 page. (If this was during when they were taking an assessment, they will not be able to retake the assessment, so the admin may need to set retake to on for the user, and provide them with an access key, in case they had already paid)



Internal Server Error

Unfortunately something went wrong on our part.

Our team has been notified and will work on resolving this as soon as possible.

We apologize for the inconvenience

- The error gets logged into AWS, if it crosses a threshold, (1 error per 5 minutes) and
- An error report is sent to the email set up by the customer.

3.7.1 Viewing errors in AWS

When the number of 500 server errors within a time frame crosses a certain threshold, an email is sent to the product-owner.

AWS Notifications Inbox - nn3un@virginia.edu February 21, 2020 at 7:06 PM AN

ALARM: "Internal Server Error" in US East (N. Virginia)

To: Nuzaba Nuzhat

<https://us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#s=Alarms&alarm=Internal%20Server%20Error>

You are receiving this email because the Alarm "Internal Server Error" in the US East (N. Virginia) region has entered the ALARM state, because its [1.0 (22/02/20 00:01:00)] was greater than the threshold (0.0) (minimum 1 datapoint for OK -> ALARM transition).

View this alarm in the AWS Management Console:
<https://us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#s=Alarms&alarm=Internal%20Server%20Error>

Alarm Details:

- Name: Internal Server Error
- Description: Oh no, a customer has faced an internal server error. Quick! Switch over to your debug email to get a grip on what's going on.
- State Change: INSUFFICIENT_DATA -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [1.0 (22/02/20 00:01:00)] was greater than the threshold (0.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Saturday 22 February, 2020 00:06:12 UTC
- AWS Account: 569902417451

You can click the link to view the error.

Alternatively you can log into your AWS account, select the cloudwatch service, and then click on alarm. Then from the left panel, select “Alarms”. Then you will see a list of alarms.

The screenshot shows the AWS CloudWatch Alarms console. The left sidebar contains navigation links: CloudWatch, Dashboards, Alarms (selected), Billing, Logs, Log groups, Insights, Metrics, Events, Rules, Event Buses, ServiceLens, and Service Map. The main panel displays a list of alarms with the following data:

Name	State	Last state update	Conditions	Actions
awseb-e-gpyaqm3jse-stack-AWSEBCloudwatchAlarmLow-1C60540C40P8E	In alarm	2020-04-08 04:56:29	NetworkOut < 2000000 for 1 datapoints within 5 minutes	-
awseb-e-gpyaqm3jse-stack-AWSEBCloudwatchAlarmHigh-2OJFQOHL7V1Y	OK	2020-04-08 04:21:10	NetworkOut > 6000000 for 1 datapoints within 5 minutes	-
Internal Server Error	Insufficient data	2020-02-26 02:00:12	500_errors > 0 for 1 datapoints within 5 minutes	-

To change subscription to the alarm, select the SNS service under AWS (<https://console.aws.amazon.com/sns/v3/home?region=us-east-1#/subscriptions>) Then from the left pane select subscriptions. From there you can delete subscriptions to specific topics, or create new ones.

The screenshot shows the Amazon SNS Subscriptions console. The left sidebar contains navigation links: Amazon SNS, Dashboard, Topics, Subscriptions (selected), Mobile, Push notifications, and Text messaging (SMS). The main panel displays a list of subscriptions with the following data:

ID	Endpoint	Status	Protocol	Topic
421e5b57-87aa-40b4-b288-9acbe110f790	arn:aws:sqs:us-east-1:569902417451:ses-bounces-queue	Confirmed	SQS	ses-bounces-topic
485f3893-7d78-4acf-bbe3-f9be1af7e797	arn:aws:sqs:us-east-1:569902417451:ses-complaints-queue	Confirmed	SQS	ses-complaints-topic
Pending confirmation	jelliottc@thesum.org	Pending confirmation	EMAIL	Default_CloudWatch_Alarms_Topic

3.7.2 What to do with errors

Log into the email address that was specified for the error reporting. (You can change this email by going to ElasticBeanstalks > Your application > Configuration > Environment Variables). There you can get a detailed report on what went wrong.

info@thepowerofdifference.org

Inbox - nn3un@virginia.edu 4:55 PM



[Django] ERROR (EXTERNAL IP): Internal Server Error: /users/4/assessments/3/statement/1

To: Nuzaba Nuzhat

Internal Server Error: /users/4/assessments/3/statement/1

DoesNotExist at /users/4/assessments/3/statement/1
Question matching query does not exist.

Request Method: GET

Request URL: <http://thepowerofdifference.org/users/4/assessments/3/statement/1>

Django Version: 2.2.6

Python Executable: /opt/python/run/venv/bin/python3

Python Version: 3.6.8

Python Path: ['/opt/python/current/app', '', '/opt/python/run/venv/local/lib64/python3.6/site-packages', '/opt/python/run/venv/local/lib/python3.6/site-packages', '/opt/python/run/venv/lib64/python3.6', '/opt/python/run/venv/lib/python3.6', '/opt/python/run/venv/lib64/python3.6/site-packages', '/opt/python/run/venv/lib/python3.6/site-packages', '/opt/python/run/venv/lib64/python3.6/lib-dynload', '/usr/lib64/python3.6', '/usr/lib/python3.6']

Server time: Sat, 11 Apr 2020 20:55:20 +0000

Installed Applications:

Unfortunately, many of the errors are quite technical in nature. Here are some solutions to some common fixes you might encounter:

1. Questions not loaded in the database: You might have forgotten to include questions.json when you deployed. In that case make sure questions.json is in your src folder, then follow the steps laid out in section 3.2
2. Question matching query does not exist: You may be missing a statement in your database, or at least one of them has the wrong question number. Go through all of them to make sure there is at least one statement with each number between 1 and 70.

If you are facing some other errors, and you have no developer to help you solve it, you should follow the following instruction:

- Go to your application environment in elasticbeanstalk and then click the Environment Actions button on the top right. From the dropdown select Restart app server(s)
- If that fails follow the steps at steps 3.3.1 to first backup the database, then follow step 3.3.3, and 3.3.2 to rebuild the environment.
- If even that fails, follow only step 3.3.3 to rebuild the site with a blank database.